

JACOB LECOQ

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PROFESSIONAL SUMMARY

Senior Software Engineer focused on platform engineering, full-stack delivery, cloud infrastructure, and AI-enabled developer workflows. Lead cross-repo modernization for Bayer Global Commerce across NestJS APIs, Nx React applications, shared UI systems, AWS Lambda ETL pipelines, Terraform infrastructure, and reusable GitHub Actions. Strong record of improving release safety, developer velocity, reporting architecture, cloud reliability, and internal automation across large enterprise systems.

WORK EXPERIENCE

Dexian (Bayer)

2024 - Present

Senior Software Engineer

- Led platform and full-stack delivery for Bayer Global Commerce, managing a production NestJS middleware API, a 22-app Nx React frontend monorepo, shared UI packages, ETL services, and reusable CI/CD automation to accelerate feature rollout and enhance system reliability
- Developed internal AI developer workflows and agent templates for ticket triage, code generation, repository and documentation retrieval, and workflow automation, reducing repetitive engineering effort and enhancing team adoption of AI-assisted development
- Automated Terraform-based AWS infrastructure for middleware services by developing modular stacks for Aurora PostgreSQL, ElastiCache Redis, DynamoDB, S3, Lambda, API Gateway, Datadog, and remote S3-backed state across non-production and production environments, reducing provisioning time and ensuring consistent environment configuration
- Delivered AWS Lambda and Nx ETL pipelines integrating SAP ERP, SAP Commerce Cloud, DynamoDB, Kafka, SQS, EventBridge, Azure hot-folder uploads, CloudWatch, Datadog, and Teams notifications to support warehouse, location, stock, and packaging-material data flows
- Integrated a deterministic workflow engine into the reporting platform by adding NestJS workflow endpoints, provider-based task execution, runtime validation, OpenAPI coverage, execution tracing, caching, and tests for packaging-material and downstream treatment reports, enabling more accurate and faster report generation
- Enhanced the React reporting platform by implementing config-driven report templates, RTK Query adapters, Redux/Zustand state management, progressive pagination, typed workflow report contracts, and reusable transformer/fetch-hook registries to improve user experience and reduce load times for large data sets
- Developed reusable GitHub Actions and composite actions for Terraform plan/apply, Docker build and publish, SonarQube scanning, npm package release, AWS authentication, and Vault-backed secret retrieval to standardize release workflows across Bayer repositories
- Modernized large TypeScript workspaces by upgrading to Node 24, TypeScript 6, Storybook 10, React ecosystem components, linting, testing frameworks, dependencies, and lockfiles to ensure continuous delivery across active product teams
- Published [@gc-agency/gc-ui](#), a standalone TypeScript component library using Vite, Storybook, Vitest, Playwright, Biome, Element Design, LaunchDarkly, and React Hook Form to standardize shared UI patterns across Bayer applications

DNAnexus

2023 - 2024

Senior Software Engineer

- Reduced AWS costs by \$30K annually through cluster optimization, pricing model adjustments, and more efficient compute orchestration
- Led backend development for production Node.js and TypeScript services, delivering three major features on schedule
- Enhanced backend quality by enforcing service boundaries, code review standards, and clearer contracts across modular services, which lowered production incidents and improved maintainability
- Designed secure cross-account AWS patterns for BYOA workloads using IAM, KMS, and Secrets Manager to enable compliant customer data processing across accounts
- Implemented Spot Instance orchestration to reduce compute costs and enhance throughput for customer workloads

Bayer

2020 - 2023

Senior Software Engineer

- Led full-stack engineering of Bayer digital platforms by developing reusable React and Node.js systems adopted across multiple programs
- Designed DynamoDB data models and indexes to enhance query performance, accuracy, and long-term maintainability
- Enhanced production reliability by implementing Datadog and CloudWatch monitoring, reducing defects, and refactoring services for improved performance
- Coordinated releases of a critical DMP application to enhance deployment reliability and enable zero-downtime delivery
- Automated AWS infrastructure using CloudFormation and Lambda, enhancing consistency and reducing manual operational tasks
- Led the migration of legacy REST endpoints to GraphQL APIs, improving data fetching efficiency and developer experience for frontend teams

Waitr

2019 - 2020

Software Engineer

- Led AngularJS frontend development for internal applications, enhancing partner and support workflows
- Optimized Grunt builds and CircleCI pipelines to reduce deployment friction and resolved production defects through targeted hot fixes

- Transformed Zeplin designs into reusable Angular components and directives, enhancing UI consistency and accelerating feature development

CGI Federal

2018 - 2019

Technical Consultant

- Collaborated with EPA stakeholders using IaC and Datadog to implement compliance-focused solutions, delivering all mandated features on schedule
- Developed full-stack EPA features using Knockout.js and Spring Boot, delivering iterative releases on a consistent two-week schedule
- Resolved cross-stack defects and redesigned service boundaries to enhance stability and long-term maintainability

ASV Global

2017 - 2018

Autonomous Systems Software Developer

- Developed C++ core systems and Qt interfaces for autonomous marine software, reducing operator steps by 33%
- Reduced regressions and accelerated defect isolation by applying code review, static analysis, QA testing, and Splunk-based log analysis
- Implemented winch-control logic, configured headless embedded systems, and calibrated instruments using Go and MATLAB, which enabled reliable field deployments without downtime

Perficient

2016 - 2017

Associate Technical Consultant

- Implemented AngularJS and Java features for government clients, enhancing release cadence and application stability
- Managed server compliance tasks and clarified requirements with clients and internal teams, reducing compliance issues
- Supported Jenkins deployments and established Git branching strategies, resulting in smoother release pipelines

EDUCATION

University of Louisiana at Lafayette

2016

Bachelor of Science, Computer Science

Lafayette, LA

CORE SKILLS

- AI-Enabled Developer Workflows, DynamoDB, GitHub Actions, Platform Engineering, Node.js, Terraform, Full-Stack Development, Cloud Infrastructure Automation, Infrastructure as Code (IaC), React, Amazon Web Services (AWS), Datadog, Redux, NestJS, Go Programming Language, Python